Acorn Additionality Assessment

[DATE OF ASSESSMENT]

Project details		
	Local partner:	Serious Shea Senegal
	Project representative:	William Kwende
	Project location - Country, Region, District(s):	Senegal, Thies, Fatick, Diourbel

1. Would farmers generally plant all the trees in one year or in phases over multiple years? (e.g. planting in groups of farmers per year)? If so, explain why?

   Farmers will plant all the trees in one year on their land. This is our recommendation, because it is easier to manage to trees, especially the fences, when they are the same age. They require the same input.

2. If planted over multiple years, for how many years does the average farmers plant trees until they reach the planned maximum density per hectare?
   - non applicable

3. What barriers did farmers face that prevented them from transitioning to a successful and long-lived agroforestry system before intervention of the local partner and Acorn?

   First of all, farmers do not know and understand agroforestery and its benefits. Growing trees is not a common activity in Sahel area, because trees don't produce income in these regions.

   a- Limited access to financial resources: Farmers in West Senegal often lack access to financial resources, which can make it difficult for them to invest in new farming practices, including agroforestry. This can also make it challenging for farmers to purchase the necessary inputs, such as seeds, tools, and equipment, required to implement an agroforestry system.

   b- Limited access to technical expertise: Many farmers in West Senegal lack the technical expertise and knowledge required to implement and maintain an agroforestry system. This includes knowledge about soil conservation, tree planting, and intercropping techniques that are necessary for successful agroforestry practices.

   c- Lack of suitable land for agroforestry: Many areas of West Senegal have degraded soils and are prone to erosion, making it challenging to implement agroforestry practices effectively. In addition, land ownership and access can also be a significant barrier for farmers, particularly for those who do not own their own land.

   d- Lack of market opportunities: Even if farmers are able to successfully implement an agroforestry system, they may struggle to find markets for their products. This can make it difficult for farmers to generate income and can discourage them from investing in new farming practices.

e- Climate variability: West Senegal experiences frequent droughts and other climate-related challenges, which can make it difficult for farmers to implement and maintain agroforestry practices over the long term.

Overall, the barriers that Senegalese farmers in West Senegal face in transitioning to agroforestry are complex and multifaceted. However, with the support of Serious Shea Senegal and organizations, farmers in West Senegal will be able to understand the benefit agroforestry and overcome most of these barriers to successfully transition towards sustainable agroforestry systems.

4. Please provide a list of proposed project activities that you offer to help farmers transition to agroforestry (e.g. agronomist advice, agroforestry training, collaboration with seed banks, site visits to successful agroforestry farms, financial support), and describe those activities.

A- Sensibilization about the benefits of agroforestry for the people and the land.

B- Farmer training and education: Providing farmers with training and education on agroforestry practices, including soil conservation, intercropping, and tree planting techniques.

C- Financial Support: Serious Shea Senegal will be providing financial support to farmers to facilitate their transition to agroforestry practices. This support will cover various aspects, including procurement, planting costs, and maintenance expenses. The financial assistance aims to alleviate the financial burden on farmers and enable them to effectively adopt agroforestry techniques.
Furthermore, farmers will have the opportunity to benefit from carbon credits. By engaging in sustainable agroforestry practices, the farmers will contribute to carbon sequestration and mitigation of greenhouse gas emissions. As a result, they will receive financial compensation through carbon credit programs, providing them with an additional source of income.
Additionally, Serious Shea Senegal plans to establish process centers in the region. Farmers will have the option to bring their harvested crops to these centers for processing. This step adds value to their agricultural produce, allowing them to access higher-value markets and generate additional income from the processed products.
Through the combination of financial support for procurement and ongoing costs, income from carbon credits, and additional financial support from the process centers, farmers will have the necessary resources and incentives to transition to agroforestry practices successfully. This holistic approach aims to promote economic sustainability and improve the livelihoods of participating farmers.

D- Nurseries with seed banks: We will establish numbers of nurseries in the planting locations. These nurseries will produce trees for the farmers.

E- Liquid Fertilisers, Liquid Pesticides: In process centers, the inclusion of biogas systems fueled by agricultural waste is a commendable initiative. These systems will generate clean energy to power the process centers, contributing to environmental sustainability and reducing reliance on conventional energy sources.

Furthermore, the biogas systems will produce highly effective liquid fertilizers as a by product. These fertilizers, derived from organic waste, provide a sustainable and nutrient-rich solution for farmers' planting and maintenance activities. By utilizing these liquid fertilizers, farmers can enhance soil fertility, promote healthy plant growth, and improve crop yields.

In addition to the utilization of organic fertilizers, your project also involves assisting and training farmers in the proper usage of pesticides. This training is essential in ensuring that farmers have the knowledge and skills to safely and effectively apply pesticides when necessary. By promoting responsible pesticide usage, you are emphasizing the importance of environmental stewardship and the protection of both crops and ecosystems.

By combining the production of clean energy, liquid fertilizers from biogas systems, and comprehensive pesticide training, your project aims to create a holistic approach to sustainable agriculture. This integrated approach benefits both the environment and farmers, promoting the use of renewable resources, improving soil health, and enhancing crop productivity while minimizing negative environmental impacts.

F- Land preparation: Preparing land for agroforestry, including soil analysis, land clearing, and erosion control measures.

G- Seed and sapling provision: Providing farmers with seeds and saplings of tree species suitable for agroforestry, including fast-growing species that can provide early benefits.

H- Tree planting and maintenance: Supporting farmers in the planting and maintenance of trees, including techniques for planting, fertilizing, pruning, and protecting trees from pests and diseases.

I- Crop diversification: Encouraging farmers to diversify their crops and integrate them with trees to improve soil health and increase yields.

J- Access to finance: Providing farmers with access to finance to invest in the necessary inputs for agroforestry, such as seeds, saplings, and tools.

K- Market development: Supporting farmers in finding markets for their products, including working with local buyers and facilitating access to larger markets.

L- Monitoring and evaluation: Regular monitoring and evaluation of the project to assess the impact and effectiveness of the program and make adjustments as needed.

5. Please select below **<u>at least one</u>** of the following barriers that would have prevented farmers from successfully transitioning to agroforestry without your support, and describe **in detail** how project interventions and the expected carbon finance will overcome this barrier?

Barrier analysis	Demonstrate that the project intervention would not have taken place due to <u>a least one</u> of the following barriers.		
Type of barrier	Examples of barrier	Situation without project	Situation with project
Financial/ economic barrier	• Insufficient financial resources to develop a project • No payment system for ecosystem services in place • No access to carbon market due to real or perceived risks associated with domestic or foreign direct investment in the country where the project intervention is to be implemented	• Insufficient financial resources to develop a project • No payment system for ecosystem services in place • No access to carbon market due to real or perceived risks associated with domestic or foreign direct investment in the country where the project intervention is to be implemented	*The project interventions, implemented by Serious Shea Senegal, aim to provide financial support to farmers, enabling them to adopt agroforestry practices. Through the project, farmers will have access to two key sources of income: carbon credits and revenue generated from the process centers. By combining income from carbon credits and revenue from process centers, the project ensures that farmers have a sustainable and viable financial model to support their transition to agroforestry. These income streams will address the financial limitations that would have otherwise prevented farmers from adopting agroforestry practices independently.* *Moreover, Serious Shea*

			Senegal's involvement and expertise in carbon credit markets and establishing process centers will facilitate the access and utilization of these financial opportunities. The project interventions aim to establish the necessary mechanisms and systems to ensure that farmers can efficiently participate in carbon markets and leverage the revenue generated from process centers. *Overall, the project interventions, supported by expected carbon finance and revenue from process centers, will overcome the financial/economic barrier by providing farmers with the necessary financial resources to successfully transition to agroforestry. This approach fosters economic sustainability, enhances farmers' livelihoods, and promotes the long-term viability of agroforestry practices.*
Technical barrier	• Lack of infrastructure for implementation of the technology • Lack of access to planting materials	• Lack of infrastructure for implementation of the technology Lack of access to planting materials	*Here's a detailed description of how project interventions will overcome this barrier:* *Lack of Infrastructure: To address the lack of infrastructure for implementing agroforestry technologies, Serious Shea Senegal will establish nurseries equipped with necessary infrastructure.*

			This includes the installation of irrigation systems, solar pumps, and other advanced technologies. These infrastructure investments will provide farmers with the necessary tools and resources to effectively grow and maintain their tree seedlings. The provision of adequate infrastructure will ensure that farmers have access to essential resources for successful agroforestry practices. *Lack of Access to Planting Materials: Another aspect of the technical barrier is the lack of access to quality planting materials. Serious Shea Senegal will play a crucial role in addressing this challenge by providing farmers with the necessary planting materials. They will ensure the availability of high-quality tree seedlings suited for agroforestry practices. By supplying farmers with the required planting materials, the project ensures that farmers have access to the essential components for their agroforestry transition.* *Additionally, the project interventions will include comprehensive training programs for farmers. These training sessions*

			will focus on the technical aspects of agroforestry, including proper planting techniques, tree maintenance, water management, and other relevant practices. By providing farmers with the necessary knowledge and skills, the project will equip them to overcome technical barriers and implement agroforestry effectively. *Through the establishment of well-equipped nurseries, provision of planting materials, and comprehensive training programs, Serious Shea Senegal will address the technical barriers faced by farmers. By overcoming these barriers, the project will enable farmers to successfully transition to agroforestry practices, ensuring the long-term viability and success of their agroforestry initiatives.*
Institutional/ political barrier	• Lack of regulations regarding agroforestry and land use management, or poor enforcement of such regulations. • Risk related to changes in government policies or laws	*<describe what institutional/political barriers (if any) are being experienced>*	*<it is not applicable for Senegal>*
Ecological barrier	• Degraded soil (e.g. water/wind erosion, salinization) • Catastrophic natural and/or human-induced events (e.g. landslides, fire) • Unfavorable meteorological conditions (e.g. early/late frost,	• Degraded soil (e.g. water/wind erosion, salinization) • Unfavorable	*For the degraded soils: Soil has been degraded due to advance of the Sahara and changing climate. The usage of bio fertilization and water retention systems will help*

	drought) • Unfavorable course of ecological successions • Biotic pressure in terms of grazing, fodder collection, etc.	meteorological conditions (e.g. early/late frost, drought)	*regenerating the soils.* *Access to Water Sources: Serious Shea Senegal will assist farmers in accessing water sources to mitigate the impact of unfavorable meteorological conditions like drought. One approach will be the establishment of deep wells, providing a reliable water supply for irrigation purposes. These wells will enable farmers to sustain their agroforestry practices even during periods of water scarcity. By ensuring access to water, the project interventions help overcome the ecological barrier posed by unfavorable meteorological conditions.* *Water Management Trainings: Alongside providing access to water sources, the project will include comprehensive water management trainings for farmers. These trainings will focus on efficient water usage techniques, such as drip irrigation, hydro retention technological systems and water conservation practices. Farmers will learn how to optimize water resources, reduce water wastage, and ensure that water is utilized effectively for agroforestry activities. By equipping*

			farmers with the knowledge and skills for water management, the project interventions will enhance their resilience in the face of ecological challenges. *Through the establishment of deep wells and water management trainings, the project interventions address the ecological barrier posed by unfavorable meteorological conditions like drought. By ensuring access to water and promoting efficient water usage, farmers will be better equipped to sustain their agroforestry practices and overcome the challenges associated with ecological constraints. This approach enhances the resilience of farmers and contributes to the long-term success of agroforestry initiatives.*
Social barrier	• Poor mobilization of local communities due to remoteness or poor infrastructure • Demographic pressure on the land (e.g. increased demand for land due to population growth) • Social conflict among interest groups in the region where the project intervention takes place • Widespread illegal practices (e.g. illegal grazing, non-timber product extraction and tree felling)	*<describe what social barriers (if any) are being experienced>*	*<Its not applicable>*
Cultural	• Lack of skilled and/or properly	• Lack of	*Here's how the project*

barrier	trained labor force • Lack of organization of local communities	skilled and/or properly trained labor force • Lack of organization of local communities	*interventions will address these barriers:* *Lack of Skilled and/or Properly Trained Labor Force: Serious Shea Senegal will play a vital role in addressing the lack of skilled and trained labor force. The project will include comprehensive training programs that cover various aspects of agroforestry, including planting, maintenance, financing, management, and pesticide usage. By providing training to the labor force, the project ensures that they acquire the necessary skills and knowledge to effectively participate in agroforestry activities. This training will enhance the capacity of the labor force, enabling them to contribute more efficiently to the implementation of agroforestry practices.* *Lack of Organization within Local Communities: The project recognizes the importance of community organization for successful agroforestry initiatives. Serious Shea Senegal, in collaboration with local partners, will work towards organizing and mobilizing local communities. This may involve establishing or strengthening existing*

			community-based organizations, cooperatives, or associations. By fostering community organization, the project aims to create a platform for collective decision-making, resource-sharing, and knowledge exchange among local farmers. This organized approach enables communities to actively participate in the planning and implementation of agroforestry practices, leading to more sustainable and impactful outcomes. *Through training programs for the labor force and the organization of local communities, the project interventions address the cultural barriers hindering the successful transition to agroforestry. By enhancing the skills and knowledge of the labor force and promoting community organization, the project empowers local stakeholders to actively engage in agroforestry activities and effectively contribute to the project's objectives. This collaborative approach ensures that cultural barriers are overcome and that the project's implementation is aligned with the needs and aspirations of the local*

			communities.